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Optical Properties of Single Molecules and One-Dimensional Chains of Organic Molecules on Surfaces

TiDE Project Proposal

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TEMPLATE-DESIGNED ORGANIC ELECTRONICS
RESEARCH TRAINING GROUP 2591



1 State of the Art

The initial proposal for the research training group Ti-DE (DFG 2551) [1] reads: “The central question is how structural order influences the optoelectronic properties of pi-conjugated materials”. The placement of this sentence at the beginning of the proposal demonstrates that this question is of paramount importance to the research group. In accordance with this question the present research project aims to obtain a thorough understanding of the relation of structural and optical properties for the example of self-organized, one-dimensional, molecular chains of linear *trans*-quinacridone on surfaces.

If transition dipole moments in fluorescent organic molecules are arranged such that they couple attractively, *J*-aggregates are formed [2]. The fluorescence of *J*-aggregates is typically red-shifted compared to the monomer. *J*-aggregates exhibit a small Stokes-shift and can exhibit novel effects, such as superradiance [3, 4] which makes them interesting for the application in optoelectronic devices such as organic light emitting diodes (OLEDs). Although many types of molecular aggregates can be classified as *J*-aggregates, these are usually three-dimensional, or even two-dimensional aggregates. The nature of molecular packing in such complicated phases leads to a variety of competing, attractive and repulsive interactions between the transition dipoles. In contrast to three-dimensional aggregates, QA forms linear, homochiral chains on metal surfaces [5–9]. A SPA-LEED image of an ordered phase consisting of linear QA-chains on Ag(100) is presented in Fig. 1. As the transition dipole moment of QA lies along the short axis of the molecule [10], these chains constitute ideal *J*-aggregates.

1.1 Preliminary Work

Fluorescence of organic molecules on bare metal surfaces is not possible due to the existence of ultrafast charge transfer mechanisms which quench the fluorescence [11]. Therefore, a thin layer of insulating materials, such as potassium chloride is deposited onto the surface [12]. The use of an insulating layer then allows for the optical analysis of fluorescent molecules such as Perylene-3,4,9,10-tetracarboxylic dianhydride (PTCDA). For PTCDA, optical spectra of dilute phases are highly resolved with the fine structure allowing an analysis of the vibrational modes [13]. Furthermore, the optical analysis yields insights on the diffusion dynamics as the fluorescence is specific to the type of adsorption site on the surface [14]. Brickwall-phases of PTCDA are also known to be superradiant [15].

Applying the insights of optical experiments with PTCDA on surfaces, on the chains of QA constitutes the main outline of the presented project.

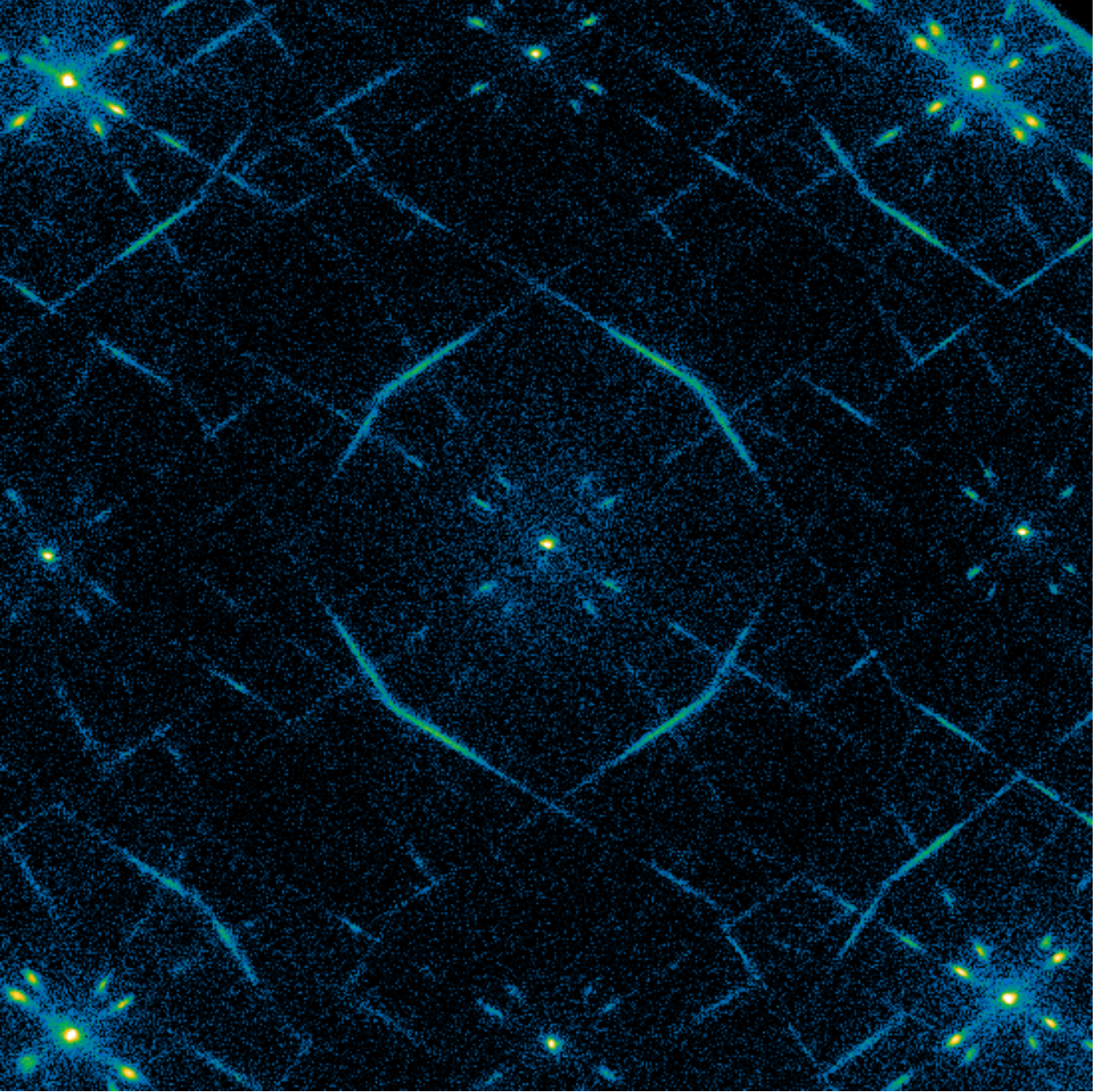


Figure 1: SPA-LEED image of the Quinacridone α -phase on Ag(100), measured at 81 eV at a substrate temperature of 75 K. The α -phase consists of linear homochiral chains of QA, as sketched in Fig. 2 on the Ag(100) surface [5].

2 Research Vision

The goal of this project is to investigate the optical properties of one-dimensional molecular aggregates of QA on surfaces and compare them to the spectra of diluted-phases, with isolated molecules. The spectra of isolated molecules on surfaces yield insights into the vibrational fine structure of the molecule.

For one-dimensional chains we expect a J-aggregate like behaviour with red-shifted fluorescence and a superradiant enhancement due to the alignment of the transition dipole moments μ , as depicted in Fig. 2 below. Ultimately we aim to determine the superradiant enhancement factor in relation to the chain-length. This offers a useful experimental comparison for established theoretical models such as the one proposed by Spano & Yamagata, in Ref. [16].

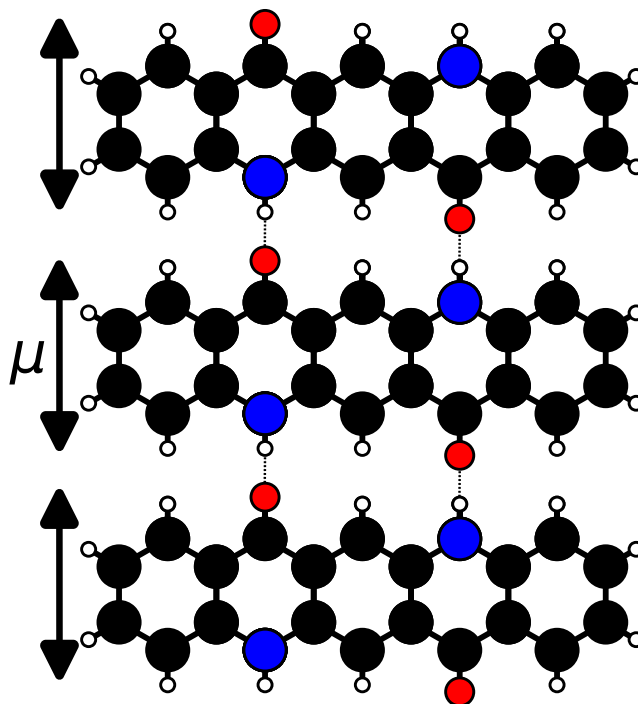


Figure 2: Schematic ball and stick model of a QA trimer, as part of linear QA-chains observed on various metal substrates[5, 6] or on thin KCl-films. The transition dipole moment μ is indicated by the arrows on the left side. Intermolecular H-bonds are indicated by the dotted lines.

2.1 Prerequisites: Spectra of QA in Solution and in Molecular Clusters on Surfaces

As a prerequisite to the presented project we aim to investigate the fluorescence of Quinacridone aggregates on the KCl/Ag(100) surface.

Measuring the fluorescence of QA of isolated molecules, linear-chains or similar phases described below, provides an experimental challenge. Therefore, the fluorescence of QA is first studied by measuring fluorescence spectra in solution (e.g., in DMSO) and for QA-clusters on surfaces. This is not just easier to perform experimentally, but serves as a useful comparison to compare future spectra with. These experiments have already concluded and some results are presented in Fig. 3. The results do not show large blue- or red-shifts in the surface spectra compared with those in DMSO solution. Furthermore, the relative peak positions align

reasonably well with spectra published in Ref. [10]. For a more in-depth comparison, fluorescence spectra with vibronic resolution are needed which comprises the next step of the project.

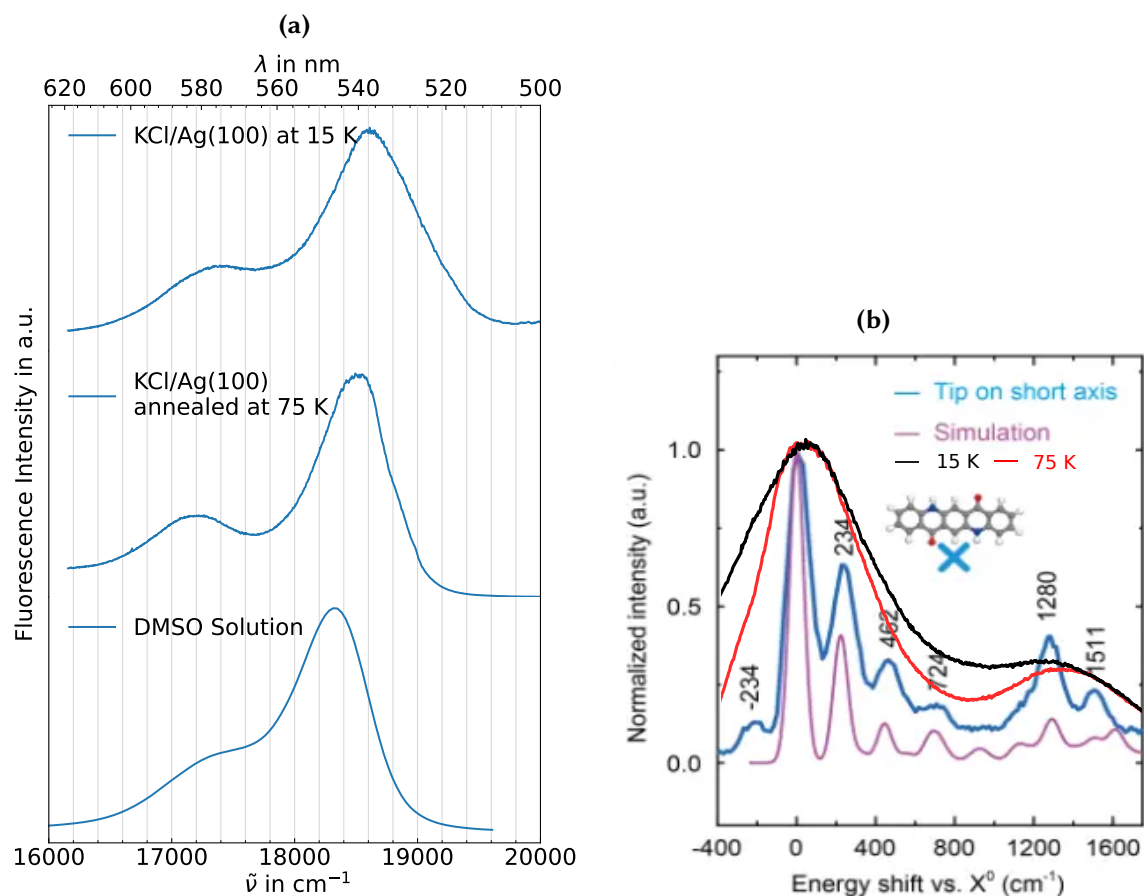


Figure 3: (a): Fluorescence spectra of Quinacridone measured on KCl/Ag(100) at 15 K, after annealing at 75 K for 5 min and in DMSO solution at 10^{-4} M. The fluorescence spectra on the surface show the same preparation of an estimated 0.05 ML of QA on $\approx 5 - 6$ ML of KCl, optically excited using a laser at an excitation wavelength of 458 nm set to 20 mW. The excitation wavelength was filtered out using a 475 nm low-pass filter before the spectrometer slit, set to 0.005 mm. (b): Fluorescence spectra shown in (a) compared with literature spectra published in [17], the literature spectra were obtained on NaCl/Ag(100) by STM-tip induced fluorescence. The simulated spectra (pink) are calculated using TD-DFT at the B3LYP/6-31G level. Taken from [17] and overlaid with own spectra.

2.2 Spectroscopy of Dilute Phases of QA on KCl/Ag(100)

In a next step, we aim to analyse the optical properties from the fluorescence of diluted-phases with isolated molecules on KCl films on the Ag(100) surface. The goal behind this step is to obtain fluorescence spectra of QA with vibronic resolution. The substrate was chosen due to established data on growing KCl films on Ag(100) [12], and due to the existence of data on linear QA-chains on that substrate [8], which is important for the later stages of this work. Comparable data of PTCDA on KCl/Ag(100) shows sharp spectral lines with a well-defined vibrational fingerprint [13]. The first aim of the project is therefore to obtain fluorescence spectra with vibronic resolution. The presence of the KCl substrate may shift the energy of the emitted light when compared to the gas-phase, as the energetic effect of the substrate on the electronic ground state and excited state may differ. However, we do not expect a significant influence on molecular vibrations compared to gas-phase spectra. As such, these experiments offer the possibility of analysing the vibrational

lines and comparing it to theoretical data. The comparison to theory allows us to confirm the reliability of theoretical workflows while deviations also yield insights into surface-modes as described in ref[13]. These surface modes may consist of frustrated rotations or translations of the molecule along the surface and are thus expected at low energetic shifts with respect to the vibrationless 0-0 transition. The presence of surface modes clarifies the bonding situation of the organic molecule on the insulator surface, such as the strength of the oxygen-surface bonds.

The structural analysis of dilute QA-phases will be performed by two complimenting methods. SPA-LEED and polarisation dependent fluorescence spectroscopy. Based on first principles calculations on LEED patterns we expect diffraction patterns of, even dilute phases, to exhibit intensity patterns close to the specular spot if the distribution of azimuthal orientations of the molecules with respect to the substrate is well defined. These types of measurements are unprecedented but may broaden the applications of SPA-LEED instruments. In order to confirm the results of these diffraction measurements we may use polarisation dependent fluorescence spectroscopy.

Once optical spectra of dilute phases with isolated molecules are measured we aim to study phase transformations induced by elevated sample temperatures, or intense optical excitation. Potentially, even at low surface coverages, different adsorption sites may lead to different spectral fingerprints [14]. The analysis of fluorescence spectra after intense optical excitation may yield insights into preferential adsorption sites.

2.3 Synthesis and Spectroscopy of Linear Molecular Chains

The main question in this thesis revolves around the optical properties of linear molecular chains of QA on KCl/Ag(100). The presence of molecular chains of QA has been reported on Ag(111), Ag(100) and Cu(111) [5, 6] in addition to thin insulating films on the latter two systems [7, 8]. As stated above, the analysis of surface fluorescence necessitates the presence of an insulating layer between metal substrates and molecule, which is why the KCl/Ag(100) system was chosen. However, the optical analysis of one-dimensional QA chains is an experimental challenge as the KCl-layer thickness needs to be fine-tuned in order to balance out two competing effects: On one side, a thicker insulating layer improves fluorescence measurement as quenching of the fluorescence is hindered. However, thinner KCl layers (2-3 ML) facilitate the synthesis of linear QA-chains, due to long-range attractive interactions between the underlying metal and QA. Adsorption of QA on thicker KCl-layers (4-6 ML) leads to the formation of aggregates with edge-on orientations which are comparable to bulk QA crystals. This can be easily rationalised when comparing the theoretical interaction energy of QA trimers on KCl(100) at 1.86 eV with sublimation enthalpies of QA crystals at 2.54 eV [8]. The aggregation of QA into bulk-like, edge-on aggregates is hence energetically favorable on thick KCl films.

In summary, the synthesis of the linear QA chains is preferentially performed on thin KCl-films, whereas fluorescence measurements work best on thick KCl-films. We aim to tackle this challenge with multiple approaches. One approach is the fine tuning of KCl layer thicknesses in order to balance the quenching of fluorescence with long-range attractive interactions of the organic molecules with the metal-substrate. Another approach is to achieve metastable linear QA-chains on thick KCl-layers by evaporation onto the substrate surface at low temperatures (≈ 10 K) followed by slow heating of the sample. Lastly replacing the metal substrate with a stepped Ag surface such as Ag(35 1 1) may induce the formation of linear chains as the QA-chains were shown to nucleate preferentially at step sites on the surface [9].

2.3.1 Optical experiments

The one-dimensional QA-chains may be seen as J -aggregates as the transition dipole moment of QA is perpendicular to the long axis of the molecule, which coincides with the direction of chain-growth [5, 17]. Therefore, the fluorescence of linear QA-chains is expected with a significant red-shift with respect to the isolated molecules. Furthermore, we expect the QA-chains to exhibit superradiance, i.e., an enhancement of the vibrationless 0-0 transition (compared to e.g. the 0-1 transition). The theoretical model for superradiance predicts a superradiant enhancement factor proportional to the coherence length N [16]. Analysing the superradiant enhancement factor in dependence of dosage or temperature, may yield further information on the length of chains, and their aggregation behaviour [15].

2.4 Additional Experiments

We note that in addition to the experiments described above, we also envision a few select experiments on QA molecules on surfaces in general. These include normal incidence x-ray standing wave (NIXSW) measurements at the Diamond light source in Oxfordshire. These aim to investigate the differences in bond-heights for the α and β phases described in Ref. [8].

2.5 Context in TiDE, and potential collaborations

Within TiDE, this project is in the **Focus Area C, Surfaces and Interfaces**. Potential collaborations with other TiDE projects may be useful for the scientific advancement of this project. Comparing spectra with theoretical calculations from Focus Area E may be useful. Also, while the outline of this project is envisioned with the one-dimensional phases of Quinacridone in mind, the investigation of QA-derivates (synthesized from members in Focus Area A) may be interesting for a deeper understanding of the underlying physico-chemical mechanisms.

3 Working Plan

In accordance with the main topics of the project described above, the working plan for the research project consists of three main milestones visualised in Fig. 4 and elaborated on below.

Milestone 1: Spectroscopy of QA in solution and in aggregates on surfaces

This milestone consists of measuring the fluorescence of QA molecules in solution in addition to starting optical measurements on the surface by investigating clusters on KCl/Ag(100). This milestone has been finished successfully.

Milestone2: Spectroscopy of Dilute QA-Phases on the surface

This milestone consists of measuring dilute QA-phases with isolated molecules on the KCl/Ag(100) surface. The main goal is to obtain fluorescence spectra with vibronic resolution. Once spectra with vibronic resolution have been successfully measured in high-quality, the aim is to study the effect of diffusion processes induced by intense light illumination or gradual annealing on the fluorescence spectra. Additionally we aim to perform SPA-LEED measurements to elucidate the structure. These experiments are performed in parallel and are planned for the latter half of 2024.

Milestone 3: Synthesis and Spectroscopy of One-dimensional QA chains on KCl/Ag(100)

The largest milestone of this project consists of analyzing optical properties of linear QA-chains on KCl/Ag(100). We consider this to be successful once superradiance has been shown in fluorescence spectra. Once the synthesis of the linear QA-chains is successful, we aim to optimise the preparation parameters in order to be able to synthesize these chains on thicker KCl-films and improve the average chain length. Following successful optimizations, we plan on studying the temperature dependence of superradiance. Similar to the first milestone, optical measurements will be performed in parallel to SPA-LEED measurements which aids us in structure determination.

Note that, the project plan also contains an unallocated three month long period. This is intentional and serves as a “buffer” period for technical projects and reparations on the vacuum chamber. At present, the plan is to perform outstanding technical repairs on the vacuum chamber after concluding the first milestone.

No specific workgroup is envisioned for the research residence at present, but the beginning of the third PhD-year is suggested as a timeslot. Currently the plan is to choose the research residence once results on linear chains exist.

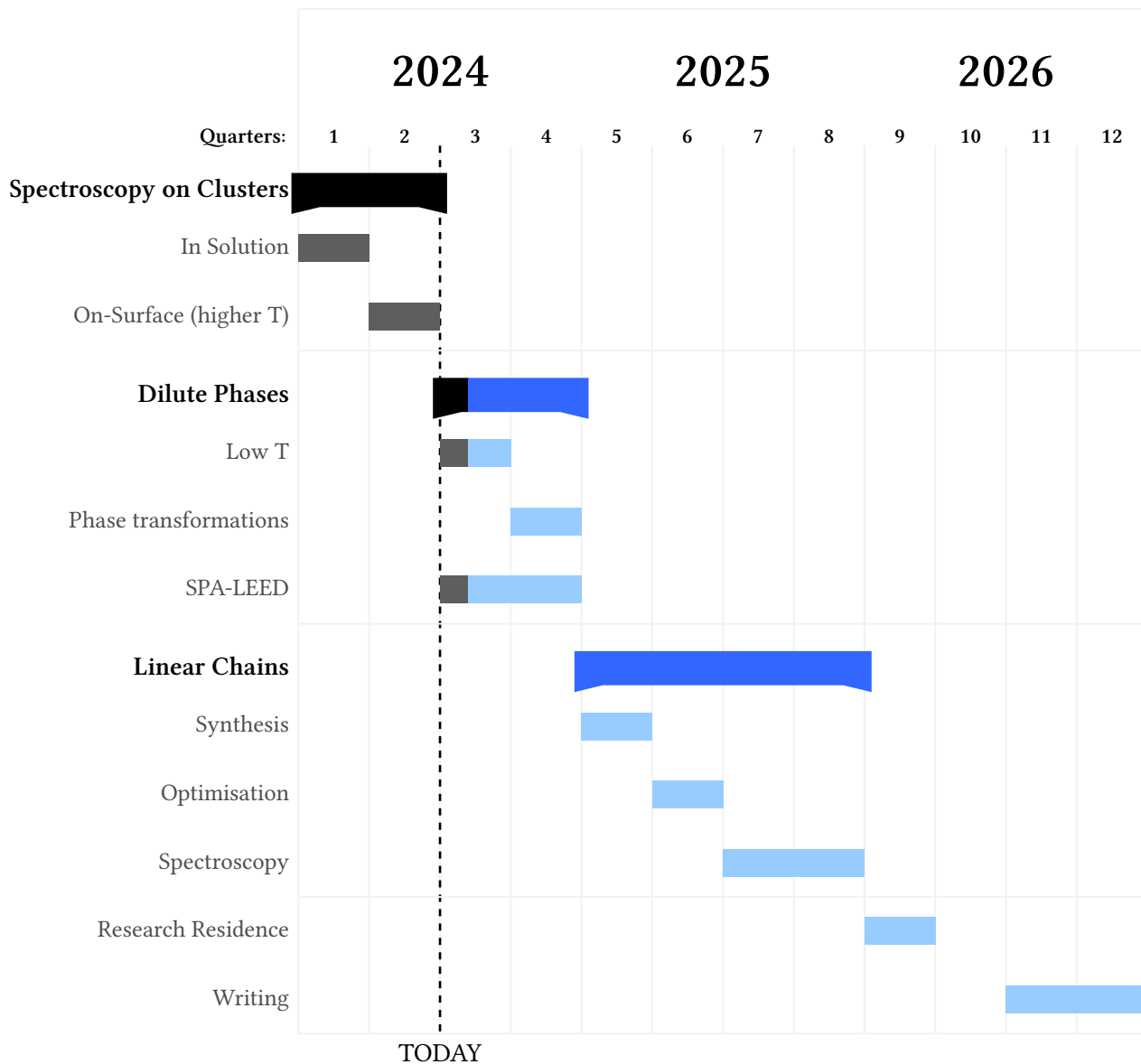


Figure 4: Gantt chart of the timeline, planned out for the presented project. Finished progress is marked in black/grey and outstanding work is depicted in blue.

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